



KTU NOTES

The learning companion.

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NOTIFICATIONS | SOLVED QUESTION PAPERS**

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CONTROL SYSTEMS

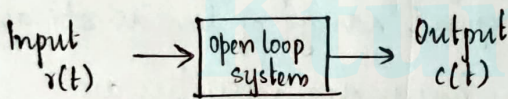
MODULE - 1

INTRODUCTION

When a number of elements or components are connected in sequence to perform a specific function, the group thus formed is called a system. In a system when the output quantity is controlled by the varying the input quantity, the system is called control system. The output quantity is called controlled variable or response and input quantity is called command signal or excitation.

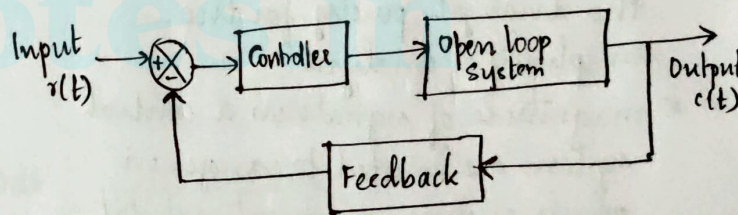
OPEN LOOP CONTROL SYSTEMS

- * system does not automatically correct the variation in its output
- * output quantity has no effect on the input quantity
- * simple and economical
- * easier to construct
- * stable systems
- * inaccurate and unreliable
- * changes in output due to external disturbances are not corrected automatically.



CLOSED LOOP CONTROL SYSTEM

- * output has an effect on the input quantity - feedback is provided
- * system is also called automatic control system.
- * systems are accurate
- * accurate even in presence of non-linearities
- * sensitivity should be made small to make system stable
- * less affected by noise
- * complex and costly
- * feedback can lead to oscillatory response
- * overall gain is reduced.

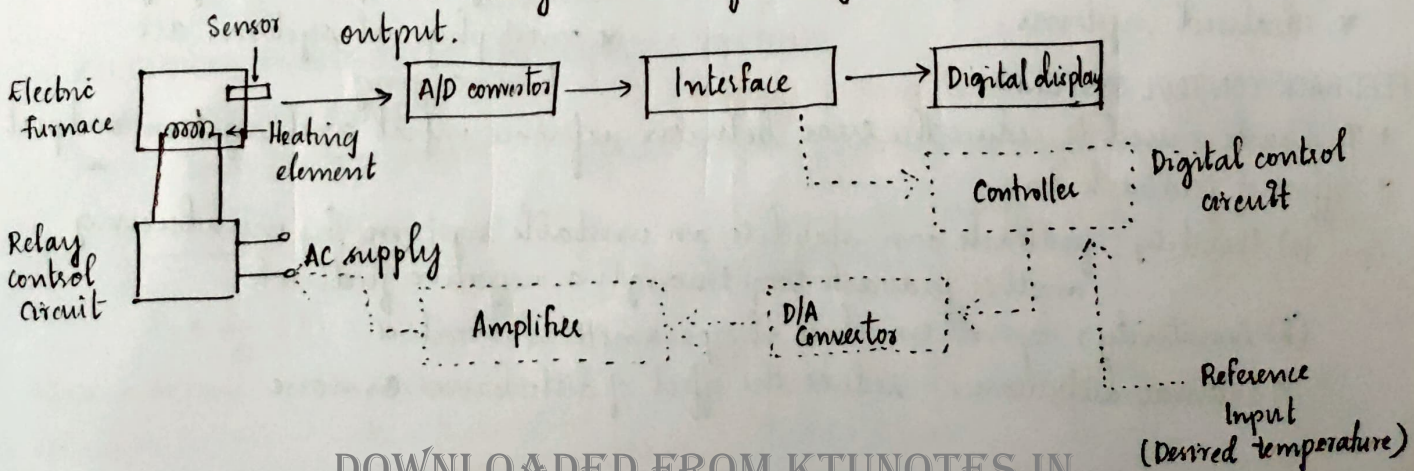


EXAMPLES OF CONTROL SYSTEMS

Example 1: Temperature control systems

Open loop system: A change in output temperature then the time setting of the relay is not altered automatically.

Closed loop system: Relay is controlled by a digital system, actual temperature is sensed and compared with desired temperature. Thus system automatically corrects any change in output.



Example 2: Traffic control system

Open loop system: traffic control by means of traffic signals operated on a time basis. The sequence of control signals are based on time slot given for each signal. System will not measure density of the traffic before giving the signals.

Closed loop system: time slots of signals are decided based on density of the traffic. Since closed loop system dynamically changes the timings, the flow of vehicles will be better than open loop system.

Example 3: Numerical Control System

Open loop system: numerical control is a method of controlling the motion of machine components using numbers. In open loop, any deviation in desired position is not checked and corrected.

Closed loop system: system automatically corrects any deviation in the desired output tool position.

LINEAR VS NON LINEAR CONTROL SYSTEMS

* Classification according to methods of analysis and design of feedback control systems.

Linear control systems

- * idealised models are fabricated by the analysts purely for the simplicity of analysis
- * magnitude of signals in a control system are limited to ranges in which system components exhibit linear characteristics

Non-linear control systems

- * magnitudes of signals are extended beyond range of linear operation
- * non linear characteristics are intentionally introduced in a control system to improve its performance or provide more effective control.

* all physical systems are non linear

TIME INVARIANT CONTROL SYSTEMS

- * parameters of a control system are stationary with respect to time during the operation of the system
- * idealised systems.

TIME VARYING CONTROL SYSTEMS

- * parameters of control system are changed with respect to time during the operation of system
- * most physical systems are time varying.

FEEDBACK CONTROL SYSTEMS

- * Feedback is used to reduce the error between reference input and system output
- * Effect of feedback on

(a) Stability: feedback can stabilise an unstable system by introducing another feedback loop through a negative feedback

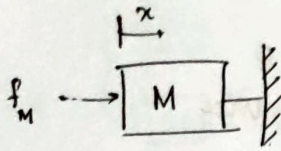
(b) Sensitivity: depends on where the parameter is located

(c) External disturbance: reduce the effect of disturbance or noise

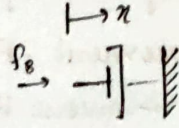
MATHEMATICAL MODELLING

Transfer function of a system is defined as the ratio of Laplace transform of output to Laplace transform of input with zero initial conditions.

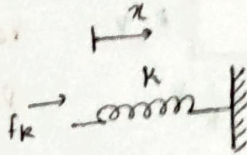
→ MECHANICAL TRANSLATION SYSTEMS



$$f_M = M \frac{d^2 x}{dt^2}$$



$$f_B = B \frac{dx}{dt}$$



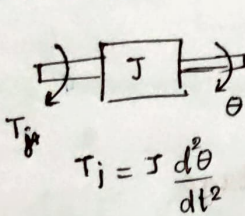
$$f_K = Kx$$

$$L[x(t)] = X(s)$$

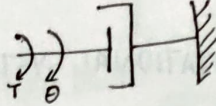
$$L\left[\frac{dx(t)}{dt}\right] = sX(s)$$

$$L\left[\frac{d^2 x(t)}{dt^2}\right] = s^2 X(s)$$

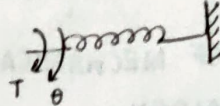
→ MECHANICAL ROTATIONAL SYSTEMS



$$T_J = J \frac{d^2 \theta}{dt^2}$$



$$T_B = B \frac{d\theta}{dt}$$



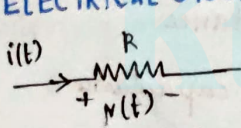
$$T_K = K\theta$$

$$L[\theta] = \Theta(s)$$

$$L\left[\frac{d\theta}{dt}\right] = s\Theta(s)$$

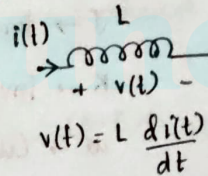
$$L\left[\frac{d^2 \theta}{dt^2}\right] = s^2 \Theta(s)$$

→ ELECTRICAL SYSTEMS



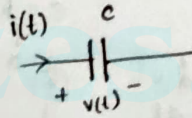
Voltage: $v(t) = Ri(t)$

Current: $i(t) = \frac{1}{R} v(t)$



$$v(t) = L \frac{di(t)}{dt}$$

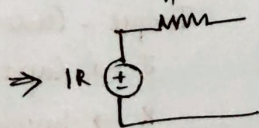
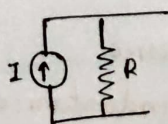
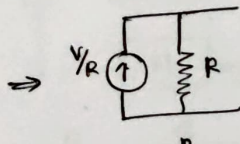
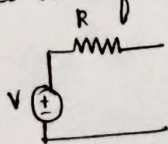
$$i(t) = \frac{1}{L} \int v(t) dt$$



$$v(t) = \frac{1}{C} \int i(t) dt$$

$$i(t) = C \frac{dv(t)}{dt}$$

Source transformation

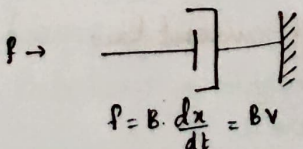


ELECTRICAL ANALOGS OF MECHANICAL SYSTEMS

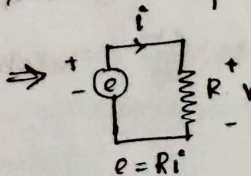
→ FORCE-VOLTAGE ANALOGY

Input: Force / Voltage source

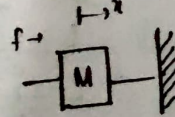
Output: Velocity / Current through element



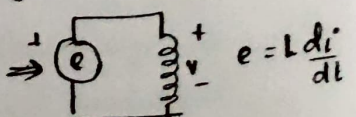
$$f = B \frac{dx}{dt} = Bv$$



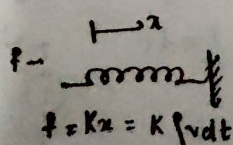
$$e = Ri$$



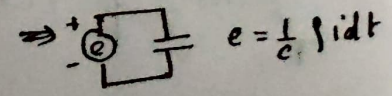
$$f = M \frac{dv}{dt}$$



$$e = L \frac{di}{dt}$$



$$f = Kx = K \int v dt$$



$$e = \frac{1}{C} \int i dt$$

Force → Voltage

B → Resistance

Velocity → Current

M → Inductance

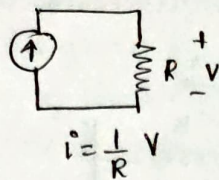
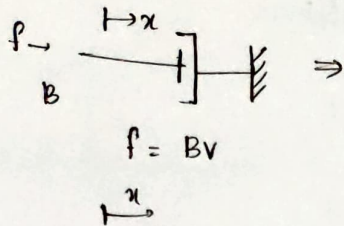
Displacement → Charge

K → Inverse Capacitance

→ FORCE CURRENT ANALOGY

Input: Force / Current

Output: Velocity / Voltage



Force - Current

Velocity - Voltage

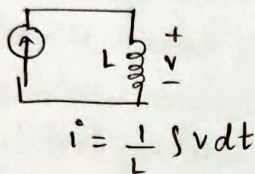
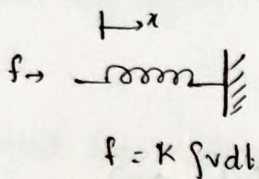
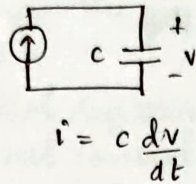
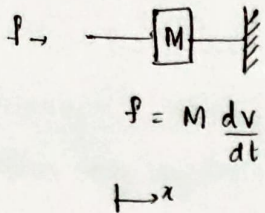
Displacement - Flux ϕ

B - Inverse Resistance

M - Capacitance

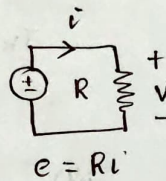
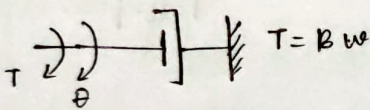
K - Inverse Inductance

Kirchoff's current law



ELECTRICAL ANALOGOUS OF MECHANICAL ROTATIONAL SYSTEMS

→ TORQUE-VOLTAGE ANALOGY



Torque - Voltage

B → Resistance

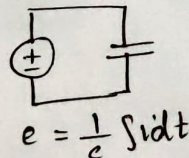
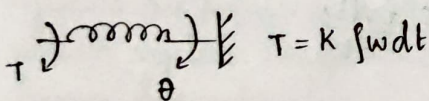
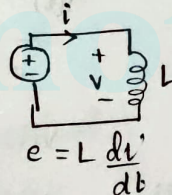
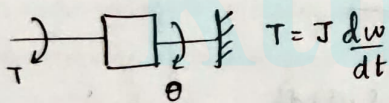
J → Inductance

K → Inverse capacitance

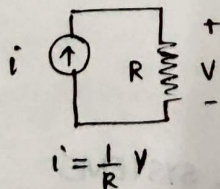
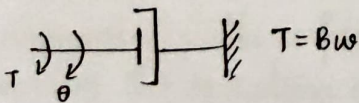
ω → Current

Angular displacement θ → charge

Kirchoff's voltage law



→ TORQUE-CURRENT ANALOGY



Torque - Current

J → Capacitance

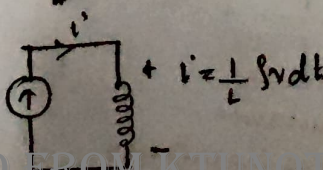
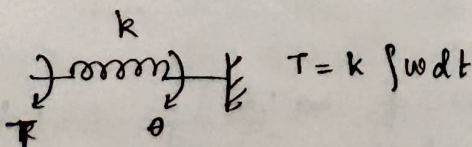
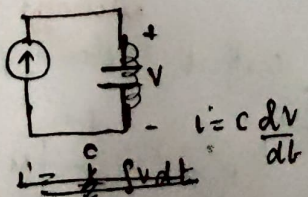
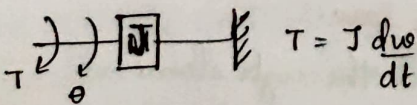
K → Inverse inductance

B → Inverse resistance

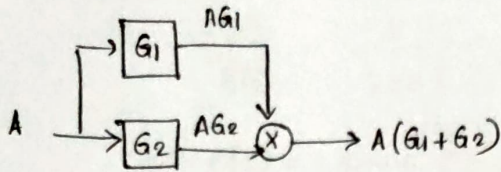
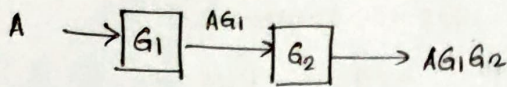
ω → Voltage

θ → Flux ϕ

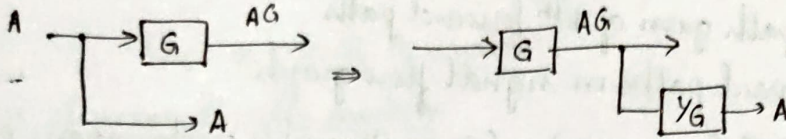
Kirchoff's current law



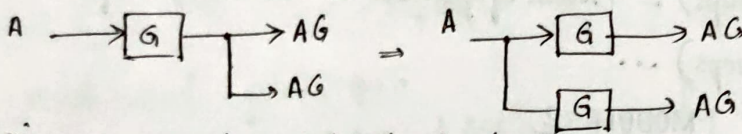
BLOCK DIAGRAM REDUCTION



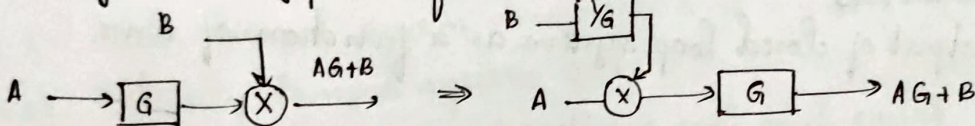
Moving branch point ahead :



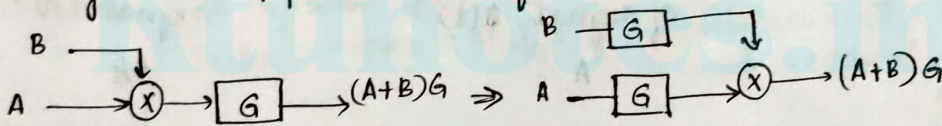
Moving branch point before block



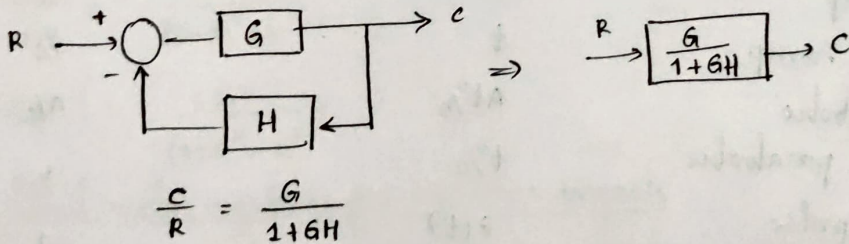
Moving summing point before block



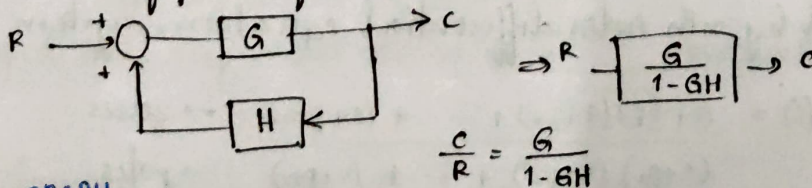
Moving summing point ahead of block



Elimination of negative (feedback loop)



elimination of positive feedback loop



SIGNAL FLOW GRAPH

Diagram that represents a set of simultaneous linear algebraic system.

Properties : * algebraic equations used must be in form of cause and effect relationship

* applicable to linear systems only

* node represents variable or signal

* branch indicates functional dependence of one signal to another

SIGNAL FLOW GRAPH REDUCTION

Mason's gain formula

$R(s)$ - Input to system $C(s)$ = Output of system

$$T(s) = \frac{C(s)}{R(s)}$$

$$\text{Overall gain } T = \frac{1}{\Delta} \sum_k P_k \Delta_k$$

$T = T(s)$ = Transfer function

P_k = Forward path gain of k^{th} forward path

k = no. of forward paths in signal flow graph

$\Delta = 1 - (\text{sum of individual loop gains}) + (\text{sum of possible combinations of two non-touching loops}) - (\text{sum of possible combinations of three non-touching loops}) \dots$

MODULE-2

TIME RESPONSE ANALYSIS

Time response - output of closed loop system as a function of time.

Test signals

Name of signal	Time domain equation of signal $g(t)$	Laplace transform of signal $R(s)$
Step	A	A/s
Unit step	1	$1/s$
Ramp	At	A/s^2
Unit ramp	t	$1/s^2$
Parabolic	$At^2/2$	A/s^3
Unit parabolic	$t^2/2$	$1/s^3$
Impulse	$\delta(t)$	1

Order of a system

If system is governed by n^{th} order differential equation, system is called n^{th} order system

$$T(s) = \frac{(s+z_1)(s+z_2) + \dots + (s+z_m)}{(s+p_1)(s+p_2) + \dots + (s+p_n)} \quad \begin{array}{l} \rightarrow \text{zeroes} \\ \rightarrow \text{poles} \end{array}$$

Partial Fraction expansion

$$T(s) = \frac{K}{s(s+p_1)(s+p_2)} = \frac{A}{s} + \frac{B}{s+p_1} + \frac{C}{s+p_2}$$

$$T(s) = \frac{K}{s(s+p_1)(s+p_2)^2} = \frac{A}{s} + \frac{B}{s+p_1} + \frac{C}{(s+p_2)^2} + \frac{D}{s+p_2}$$

$$T(s) = \frac{K}{(s+p_1)(s^2+bs+c)} = \frac{A}{s+p_1} + \frac{Bs+C}{s^2+bs+c}$$

Response of first order system

$$\frac{C(s)}{R(s)} = \frac{1}{1+sT}$$

$$c(t) = 1 - e^{-t/T} \quad (\text{Unit step input})$$

$$c(t) = A [1 - e^{-t/T}] \quad (\text{step response})$$

$T \rightarrow$ time constant of system

Response of second order system

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

ω_n - undamped natural frequency

ζ - Damping ratio

CASE 1: Undamped system $\zeta=0$

$s_1, s_2 = \pm j\omega_n$ } roots are purely imaginary

CASE 2: Under damped system $0 < \zeta < 1$

$s_1, s_2 = -\zeta\omega_n \pm j\omega_d$ } roots are complex conjugate

CASE 3: critically damped system $\zeta=1$

$s_1, s_2 = -\omega_n$ } roots are real and equal

CASE 4: Overdamped system $\zeta > 1$

$s_1, s_2 = -\zeta\omega_n \pm \omega_n \sqrt{\zeta^2 - 1}$ } roots are real and unequal

$$\mathcal{L}\{1\} = \frac{1}{s}$$

$$\mathcal{L}\{e^{-at} \sin \omega t\} = \frac{\omega}{(s+a)^2 + \omega^2}$$

$$\mathcal{L}\{e^{-at} \cos \omega t\} = \frac{s+a}{(s+a)^2 + \omega^2}$$

Response of second order systems for various inputs

undamped unit step input $\rightarrow 1 - \cos \omega_n t$

step input $\rightarrow A(1 - \cos \omega_n t)$

underdamped unit step input $\rightarrow 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} \sin(\omega_d t + \theta) \quad \theta = \tan^{-1} \frac{\sqrt{1-\zeta^2}}{\zeta}$

critically damped unit step $\rightarrow 1 - e^{-\omega_n t} (1 + \omega_n t)$

over damped unit step $\Rightarrow 1 - \frac{\omega_n}{2\sqrt{\zeta^2-1}} \left[\frac{e^{-s_1 t}}{s_1} - \frac{e^{-s_2 t}}{s_2} \right]$

$$s_1 = \zeta\omega_n - \omega_n \sqrt{\zeta^2 - 1}$$

$$s_2 = \zeta\omega_n + \omega_n \sqrt{\zeta^2 - 1}$$

TIME DOMAIN SPECIFICATIONS

(a) Rise time - time taken by response to rise from 0% to 100%

$$t_r = \frac{\pi - \theta}{\omega_d} \quad \theta = \tan^{-1} \frac{\sqrt{1-\zeta^2}}{\zeta}$$

$$\omega_d = \omega_n \sqrt{1-\zeta^2}$$

(b) Peak time - time taken by response to reach the peak value

$$t_p = \frac{\pi}{\omega_d} \quad \omega_d = \omega_n \sqrt{1-\zeta^2}$$

(c) Peak overshoot M_p - ratio of maximum peak value to final value

$$M_p = e^{-\frac{\pi \zeta}{\sqrt{1-\zeta^2}}} \quad M_p = \frac{c(t_p) - c(\infty)}{c(\infty)}$$

(d) Settling time - time taken by response to reach and stay within a specified error (2% or 5%)

$$t_s = \frac{4}{\zeta \omega_n} = \frac{\ln(\% \text{ error})}{\zeta \omega_n}$$

STEADY STATE ERROR

Type number of control systems

Given by number of poles of loop transfer function at origin - decides the steady state error.

Type number	Order
* specified for loop transfer function	* specified for any transfer function
* given by no. of poles lying at origin of s plane	* given by no. of poles of transfer function

Steady state error

$$e_{ss} = \lim_{s \rightarrow 0} s \cdot \frac{R(s)}{1+G(s)}$$

$$e_{ss} = \lim_{s \rightarrow 0} s E(s)$$

$$E(s) = \frac{c(s)}{G(s)}$$

Steady state error for various inputs

	0	1	2	3	← Type no. of system
Unit step	$\frac{1}{1+K_p}$	0	0	0	
Unit ramp	0	$\frac{1}{K_v}$	0	0	
Unit parabolic	0	0	$\frac{1}{K_a}$	0	

Static error coefficients

K_p - positional error coefficient $K_p = \lim_{s \rightarrow 0} G(s)H(s)$ Type 0

K_v - velocity error coefficient $K_v = \lim_{s \rightarrow 0} s G(s)H(s)$ Type 1

K_a - acceleration error coefficient

FREQUENCY DOMAIN SPECIFICATIONS

1. Resonant Peak M_p

maximum value of magnitude of closed loop transfer function

for second order system $M_p = \frac{1}{2\zeta\sqrt{1-\zeta^2}}$

2. Resonant Frequency ω_r

frequency at which resonant peak occurs

$$\omega_r = \omega_n \sqrt{1-2\zeta^2}$$

3. Bandwidth ω_b

range of frequencies for which system's normalised gain is more than -3dB.

$$\omega_b = \omega_n \left[1-2\zeta^2 - \sqrt{2-4\zeta^2+4\zeta^4} \right]^{1/2}$$

4. Cut off rate

slope of log-magnitude curve near the cut off frequency

5. Gain margin K_g

value of gain to be added to system in order to bring the system to the verge of instability.

$$K_g \text{ in dB} = \frac{1}{|G(j\omega_p)|}$$

ω_p - phase crossover frequency

6. Phase margin γ

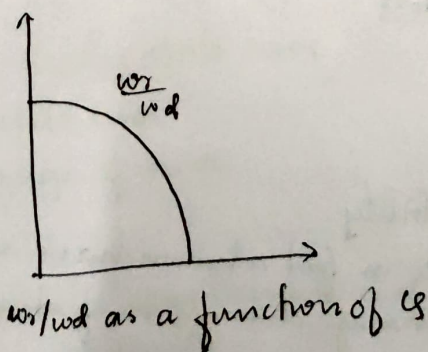
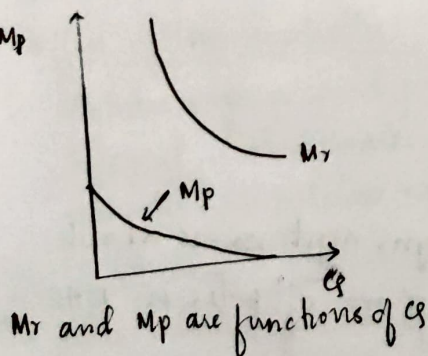
$$\gamma = 180^\circ + \phi_{gc}$$

ϕ_{gc} - gain crossover frequency

$$\gamma = 90^\circ - \tan^{-1} \left[\frac{(-2\zeta^2 + \sqrt{4\zeta^4 + 1})^{1/2}}{2\zeta} \right]$$

CORRELATION BETWEEN TIME AND FREQUENCY RESPONSE

M_r, M_p



CONCEPT OF BIBO STABILITY

Bounded input - bounded output - system is stable

If all roots lie on left of s -plane, then response $m(t)$ decreases to zero
 $\therefore$ System is stable

If any root lie on right of s -plane, system is unstable

If there are one or more repeated roots on $j\omega$ axis, system is marginally stable

Absolute stability: system output is stable for all variations of its parameters.

$$\text{Response } c(t) = \int_0^{\infty} m(\tau) r(t-\tau) d\tau$$

$$|c(t)| = \left| \int_0^{\infty} m(\tau) r(t-\tau) d\tau \right|$$

$$|c(t)| \leq \int_0^{\infty} |m(\tau) r(t-\tau)| d\tau$$

$$\leq \int_0^{\infty} |m(\tau)| |r(t-\tau)| d\tau$$

$$\int_0^{\infty} |m(t)| dt < \infty - \text{for system to be stable.}$$

ROUTH - HURTWIZ CRITERION

$$a_0 s^n + a_1 s^{n-1} + a_2 s^{n-2} + a_3 s^{n-3} + \dots + a_n s^0 = 0$$

$$s^n \quad a_0 \quad a_2 \quad a_4 \quad \dots \quad a_n$$

$$s^{n-1} \quad a_1 \quad a_3 \quad a_5$$

$$s^{n-2} \quad \frac{a_1 a_2 - a_0 a_3}{a_1}$$

;

$$s^0 \quad a_n$$

Conditions for stability

- ① All coefficients in first column have same sign, system is stable
- ② No. of sign change changes in first column is no. of poles in RHS of s -plane

* If there is a '0' term in first column - replace it with ϵ .

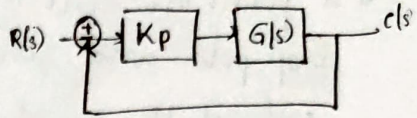
* If there is an all zero row, form auxiliary equation with row just above row.

EFFECT OF P, PI & PD CONTROLLERS

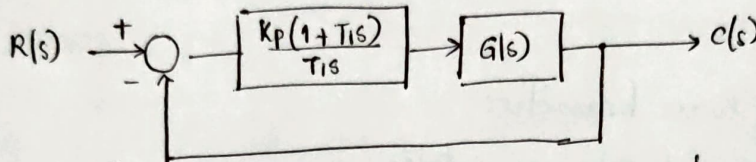
Effect of P (proportional) controller

Loop gain $\uparrow$ Sensitivity $\downarrow$

Proportional controller produces an output proportional to error signal. Transfer function of proportional controller is K_p . Proportional amplifier amplifies error signal and increases loop gain of the system. It decreases the sensitivity of system to parameter variations.



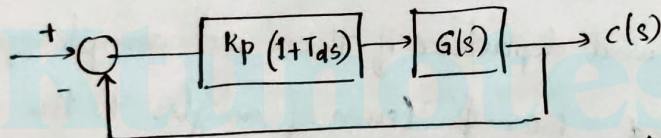
Effect of PI (proportional integral) controller



Order increase by one
 $\hookrightarrow$ less stable system
 $\hookrightarrow$ Type no. increase
reduce steady state error

Output signal consists of two terms - one proportional to error signal and one proportional to integral to error signal. PI controller introduces a zero in the system and increase order by one. Increase of order results in less stable system. PI controller increases the type number. Results in reducing steady state error.

Effect of PD (proportional derivative) controller



Introduces a zero
Damping ratio $\uparrow$
Peak overshoot $\uparrow$
Rise time $\downarrow$

Output signal consists of two terms - one proportional to error signal and one proportional to derivative of error signal. PD controller introduces a zero in system and increases the damping ratio. This increase peak ~~sat~~ overshoot and reduce the rise time. PD controller does not modify the type number.

Effect of PID controller

Stabilises gain and produces a steady state error.
Integral - eliminates steady state error
Derivative - reduces rate of change of error

ROOT LOCUS

The path taken by roots of a characteristic equation when open loop gain K is varied from 0 to ∞ is called as root locus.

Angle and magnitude condition

$$|G(s) \cdot H(s)| = 1$$

$$\angle G(s) \cdot H(s) = 180^\circ (2q+1)$$

Properties and construction of root locus

- * root locus starts from open loop poles for $k=0$ and ends at open loop zeros for $k=\infty$.
- * root locus is symmetrical to real axis
- * a point on real axis is part of root locus iff the total no of open loop poles and zeros on the real axis lie to the right hand side of the point is an odd number

Construction of root locus:

- 1) Find real axis root locus branches
- 2) Find centroid and angle of asymptotes

$$\theta = \frac{180}{P-Z} (2q+1) \quad \Leftarrow \text{Angle of asymptotes} \quad q = 0, 1, 2, 3, \dots$$

$$N = P - Z \quad \Leftarrow \text{Total no. of asymptotes}$$

$$\text{Centroid } \sigma = \frac{\sum P_R - \sum Z_R}{P - Z}$$

- 3) Find angle of arrival and departure if there are complex zeros or poles

Angle of departure $\phi_d = 180 - \phi$ (sum of angles to the complex poles from other poles - sum of angles to the complex pole from other zeros)

angle of arrival

$$\phi_a = 180 + \phi$$

- 4) Find the break point

$$\text{angle of break point} = \pm \frac{180}{n} \quad n \rightarrow \text{no. of poles meeting at breakpoint}$$

To find break point

(a) Equate $G(s) \cdot H(s) = -1$ and express k as function of s

(b) Find $\frac{dk}{ds}$ and equate to zero

(c) let s_1, s_2 be roots. If substituting in characteristic equation gives real and positive value of k

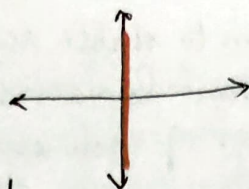
- 5) If the asymptotes cross $j\omega$ axis, find value of k at which root locus intersects with $j\omega$ axis.

$\Rightarrow$ substitute $s = j\omega$ in characteristic equation.

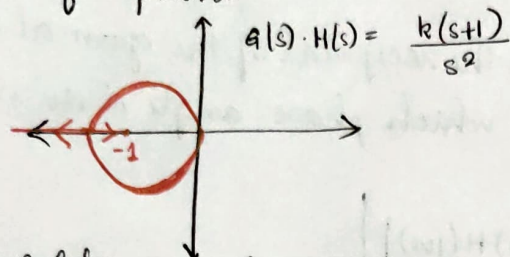
$$\alpha = \cos^{-1} \zeta \quad [\text{underdamped system}]$$

INTRODUCING POLES AND ZEROES - EFFECT

Consider $G(s) \cdot H(s) = k/s^2$

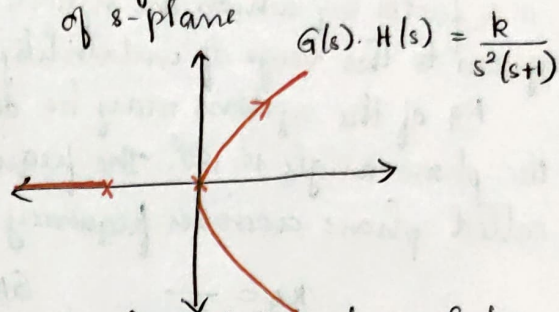


Introducing a zero to left of s-plane



- Stability increases
- Settling time decreases

Introducing a ~~zero~~ pole to right left of s-plane



- Repels root locus branch to right of s plane
- stability decreases, settling time ↑

MODULE - 4

NYQUIST STABILITY CRITERIA

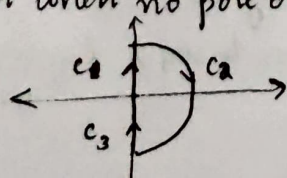
Principle of argument: Let $F(s)$ is a single valued rational function and is analytic in a given region in the s-plane except at some points. For every point on the closed contour on the s-plane, the corresponding $F(s)$ plane contour will encircle the origin N times in anticlockwise direction where N is the difference between no. of poles and no. of zeroes of $F(s)$ that are enclosed by chosen closed contour in s-plane.

- * If N is positive direction in $F(s)$ plane - anticlockwise
- If N is zero, no encirclement of origin
- If N is negative, direction in $F(s)$ plane - clockwise

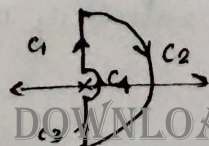
Nyquist Stability criteria

If the $G(s)H(s)$ contour in the $G(s)H(s)$ plane corresponding to Nyquist contour in the s-plane encircles the point $-1+j0$ in the clockwise direction as many times as the no. of right half s-plane poles of $G(s)H(s)$, then the closed loop system is stable.

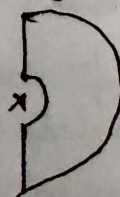
(a) Nyquist contour when no pole on imaginary axis



(b) Nyquist contour when there are poles at origin



(c) Nyquist plot with poles at origin and imaginary axis



Relative stability

indicates the closeness of the system to stable region. Indication of the strength or degree of stability. In time domain, relative stability measured by relative settling times of each root or pair of roots

Gain margin (K_g)

is a factor by which the system gain can be increased to drive the system to the verge of instability

K_g of the system may be defined as the reciprocal of the gain at which the phase angle is 180° . The frequency at which phase angle is 180° is called phase crossover frequency

$$K_g = \frac{1}{G_A} \quad G_A = |G(j\omega)H(j\omega)| \Big|_{\omega=\omega_{pc}}$$

$$K_g \text{ in dB} = -20 \log G_A$$

Phase margin γ

amount of additional phase lag at gain crossover frequency required to bring the system to verge of instability.

$$\text{Phase margin } \gamma = 180^\circ + \phi_{gc}$$

STABILITY ANALYSIS USING BODE PLOT

For stability of closed loop system, the gain margin of open loop system should be greater than 1 or if it is expressed in dB, it should be positive and phase margin of open loop system should be positive.

Bodeplot

Procedure for magnitude plot

1) Convert transfer function into bode form or time constant form
write terms in $1+sT$ form and give $s = j\omega$.

2) List the corner frequencies in increasing order and prepare a table

Term	Corner Frequency	Slope	Change in slope
------	------------------	-------	-----------------

3) Consider value of $k=1$ or $k=?$ at ω_L (lowest value)

4) Calculate gain in dB at every frequency (corner)

$$\text{Gain at } \omega_y = \left[\text{slope change from } \omega_x \text{ to } \omega_y \times \log \frac{\omega_y}{\omega_x} \right] + \text{Gain at } \omega_x$$

5) Mark in semilog sheet

Procedure for phase plot

1) Calculate phase angles of $G(j\omega)$ for various values of ω

2) Draw in semilog sheet

STATE VARIABLES

Linearly independent system variable, completely determine the value of all system variable for $t \geq t_0$

System variable - any variable that responds to an input as initial condition in a system.

State vector - a vector whose elements are state variable

State space - n dimensional space vector whose axis are state variables

STATE EQUATIONS

algebraic equations that express combination of state variables and inputs

$$\dot{x}(t) = Ax(t) + Bu(t) \quad \text{— state equation}$$

$$y(t) = Cx(t) + Du(t) \quad \text{— output equation}$$

OR

$$\dot{x} = Ax + Bu \quad y = Cx + Du \quad t \geq t_0$$

initial conditions $x(k+1) = G(k)x(k) + H(k)u(k)$

$$y(k) = c(k)x(k) + D(k)u(k)$$

x - state variable

$\dot{x}$ - derivative w.r.t t

y - output vector

u - control vector

A - System matrix

B - Input matrix

C - Output matrix

D - Scalar Quantity

STATE SPACE REPRESENTATION - MECHANICAL SYSTEM

$$u(t) = M \frac{d^2 y(t)}{dt^2} + b \frac{d}{dt} y(t) + ky(t)$$

$$U(s) = Ms^2 Y(s) + Bs Y(s) + kY(s)$$

$$U(s) = M\ddot{y} + b\dot{y} + ky \quad (1)$$

State variable

$$x_1 = y \quad x_2 = \dot{y} \quad x_3 = \ddot{y}$$

$$x_2 = \dot{x}_1$$

$$x_3 = \dot{x}_2$$

$$\ddot{y} = \frac{U - b\dot{y} - ky}{M}$$

$$\dot{x}_2 = \frac{U - bx_2 - kx_1}{M}$$

$$\dot{x}_2 = \frac{U}{M} - \frac{b}{M}x_2 - \frac{k}{M}x_1$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -k/M & -b/M \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1/M \end{bmatrix} U$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

STATE SPACE REPRESENTATION - ELECTRICAL SYSTEM

$$e_i(t) = L \frac{di(t)}{dt} + Ri(t) + \frac{1}{C} \int i(t) dt$$

$$E_i(s) = LsI(s) + RI(s) + \frac{1}{Cs} I(s)$$

$$= \left[\frac{Ls^2 + RCs + 1}{Cs} \right] I(s)$$

$$\int i(t) dt = \frac{1}{s} i(s)$$

$$\frac{di(t)}{dt} = s i(s)$$

$$I(s) = \frac{sC Ei(s)}{s^2LC + RCs + 1}$$

$$e_o(s) = \frac{1}{C} \frac{i(s)}{s}$$

$$e_o(s) = \frac{sC Ei(s)}{s^2LC + RCs + 1} \cdot \frac{1}{Cs} = \frac{Ei(s)}{s^2LC + sRC + 1}$$

$$s^2LC e_o(s) + sRC e_o(s) + e_o(s) = Ei(s)$$

$$LC \ddot{e}_o + RC \dot{e}_o + e_o = Ei$$

$$\text{let } x_3 = \ddot{e}_o \quad x_2 = \dot{e}_o \quad x_1 = e_o \quad x_2 = x_3$$

$$LC \dot{x}_2 + RC x_2 + x_1 = Ei$$

$$\dot{x}_2 = \frac{1}{LC} Ei - \frac{R}{L} x_2 - \frac{1}{LC} x_1$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1/LC & -R/L \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1/LC \end{bmatrix} Ei$$

$$e_o = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

TRANSFER FUNCTION

$$\dot{x} = AX + BU$$

$$y = CX + DU$$

$$sX(s) = AX(s) + BU(s)$$

$$Y(s) = CX(s) + DU(s)$$

$$X(s) [sI - A] = BU(s)$$

$$Y(s) = B(sI - A)^{-1} U(s) + D U(s)$$

$$Y(s) = \underbrace{[C(sI - A)^{-1} B + D]}_{\text{Transfer function matrix}} U(s)$$

Pulse transfer function

$$G(s) = C(sI - A)^{-1} B + D$$

To find inverse of a matrix

$$A^{-1} = \frac{\text{adj } A}{|A|}$$

$$\text{adj } A = [\text{co-factor matrix of } A]^T$$

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \quad \text{cofactor matrix of } A = \begin{bmatrix} ei - fh & -(di - fg) & dh - eg \\ -(bi - ch) & ai - fg & -(ah - bg) \\ bf - ec & -(af - ed) & ae - bd \end{bmatrix}$$

$$|A| = a(ei - fh) - b(di - fg) + c(dh - eg)$$

Method 1: Laplace transform method

$$\dot{X} = AX + BU$$

$$Y = CX + P$$

$$sX(s) = AX(s) + B U(s)$$

$$X(s) = [sI - A]^{-1} x(0) + [sI - A]^{-1} B U(s)$$

$$x(t) = \underbrace{L^{-1} \{ [sI - A]^{-1} x(0) \}}_{\text{zero input response}} + \underbrace{L^{-1} \{ [sI - A]^{-1} B U(s) \}}_{\text{zero state response}}$$

$$\text{Let } \phi(t) = L^{-1} \{ [sI - A]^{-1} \}$$

$$x(t) = \phi(t) x(0) + L^{-1} \{ \phi(s) B \cdot U(s) \}$$

Method 2: Infinite series method

$$\dot{X} = AX + BU$$

if input $u=0$

$$\dot{X} = AX \leftarrow \text{solution gives ZIR}$$

$$x(t) = e^{At} x(0)$$

$$\text{Similarly } \bar{x}(t) = e^{At} \bar{x}(0)$$

STATE TRANSITION MATRIX

When input $u=0$, $x(t) = \phi(t) \cdot x(0)$

$$\phi(t) = e^{At}$$

$$\text{State transition matrix } \phi(t) = L^{-1} \{ [sI - A]^{-1} \} = I + At + \frac{A^2 t^2}{2!} + \dots$$

Properties of STM

1. $\phi(0) = I$
2. $\phi^{-1}(t) = \phi(-t)$
3. $\phi(t_1 + t_2) = \phi(t_1) \cdot \phi(t_2)$
4. $[\phi(t)]^k = \phi(kt)$

CONTROLABILITY AND OBSERVABILITY

A system is said to be controllable if it is possible to transfer the system from any given initial state $x(t_0)$ to any given final state $x(t)$ in a finite time interval by changing the input u .

Kalman's test for controllability

$$Q_c = [B \quad AB \quad A^2B \quad \dots \quad A^{n-1}B]$$

A system is controllable if Q_c has rank $= n$

If Q_c is a square matrix of order $n \times n$, then Q_c has rank n iff $|Q_c| \neq 0$

A system is said to be observable if it is possible to determine the initial states of the system in a finite time interval by observing the output

Kalman's test for observability

$$Q_o = \begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \\ CA^{n-1} \end{bmatrix}$$

A system is observable if rank of $Q_o = \text{Rank of } A$

i.e. $|Q_o| \neq 0$